

SEMICONDUCTOR ELECTRONICS :

MATERIALS, DEVICES AND SIMPLE CIRCUITS

Semiconductor

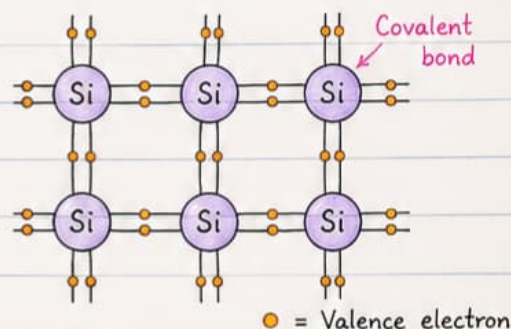
- Materials whose conductivity lies between conductor and insulator are called **semiconductors**.
- Examples** : Silicon (Si), Germanium (Ge), Gallium arsenide (GaAs) etc.
- At room temperature, semiconductor has **moderate conductivity** due to both **electrons** and **holes**.

CONDUCTIVITY COMPARISON

Low			High
Insulator (e.g. Glass) $\sim 10^{-10}$ S/m		Semiconductor (e.g. Si) $\sim 10^{-4}$ S/m	Conductor (e.g. Cu) $\sim 10^7$ S/m

INTRINSIC SEMICONDUCTOR

- A pure semiconductor is called **intrinsic** semiconductor.
- In pure Si or Ge, each atom forms four **covalent bonds** with its neighbours.
- At absolute zero (0 K), there are no free electrons.
- At room temperature, some electrons gain **thermal energy** and jump from **valence band** to **conduction band** leaving behind **holes**.



ENERGY BAND DIAGRAM (Intrinsic Semiconductor)

Conduction Band

 E_g (Energy gap)

Valence Band

- Electrons in valence band are bound.
- Electrons in conduction band are free to move.
- E_g (band gap) for Si = 1.1 eV
for Ge = 0.7 eV

Key Point

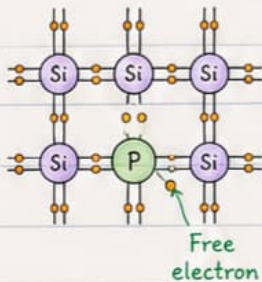
In intrinsic semiconductor at room temperature, number of electrons (n) = number of holes (p)

EXTRINSIC SEMICONDUCTOR

- A small amount of impurity is added to pure semiconductor to change its conductivity. There are two types :

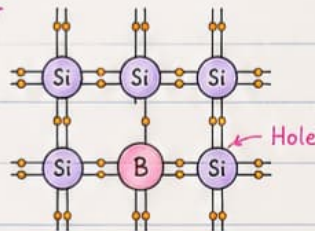
(i) n-Type Semiconductor

- Doped with pentavalent impurity (e.g. P, As, Sb).
- Four electrons form bonds, fifth electron is free.
- Majority carriers $\rightarrow$ electrons
- Minority carriers $\rightarrow$ holes



(ii) p-Type Semiconductor

- Doped with trivalent impurity (e.g. B, Al, Ga).
- Three electrons form bonds, one bond incomplete $\rightarrow$ hole is created.
- Majority carriers $\rightarrow$ holes
- Minority carriers $\rightarrow$ electrons



APPLICATIONS OF SEMICONDUCTORS

LED
(Light Emitting Diode)Solar Cells
(Photovoltaic Effect)Transistors
(Amplifiers, Switches)Diodes
(Rectification)Integrated Circuits
(Used in all electronic devices)

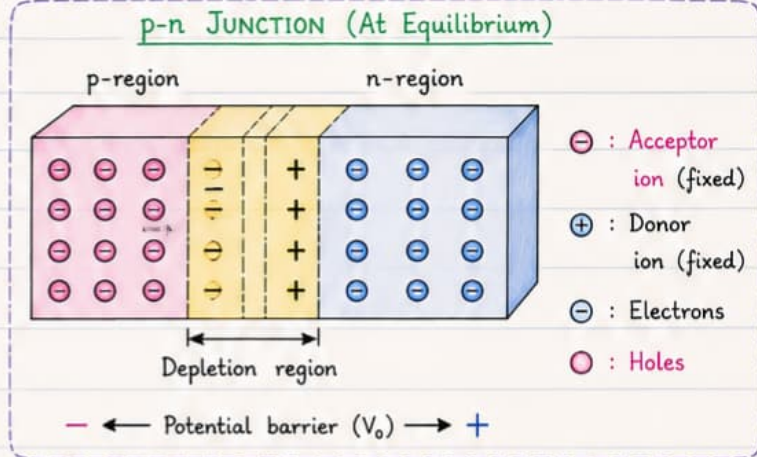
NEET TIP

Focus on energy band diagram, n-type & p-type formation, and working of p-n junction diode.

THE p-n JUNCTION DIODE

1. FORMATION OF p-n JUNCTION

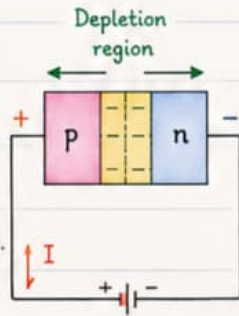
- When p-type and n-type semiconductors are joined, **electrons** diffuse from n-side to p-side and **holes** from p-side to n-side.
- This diffusion leaves behind fixed ions and creates a **depletion region** near the junction.
- A **potential difference** is set up across the junction which opposes further diffusion.



2. BIASING OF p-n JUNCTION DIODE

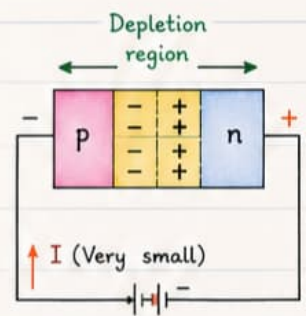
(i) FORWARD BIAS

- p-side connected to +ve terminal of battery and n-side to -ve.
- Potential barrier decreases.
- Depletion width decreases.
- Majority carriers cross the junction.
- Large current flows.

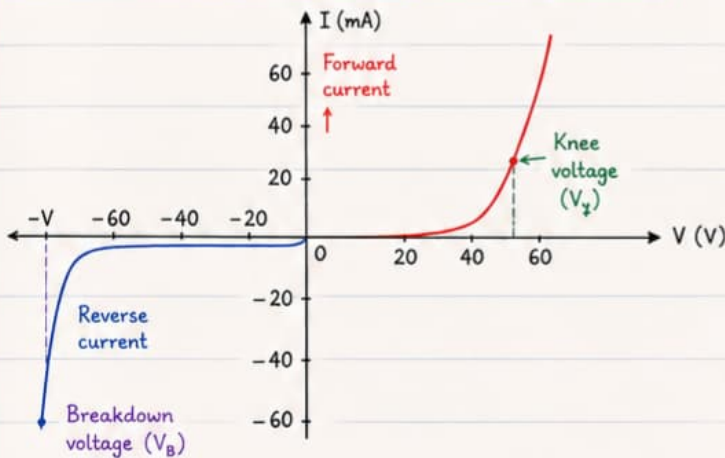


(ii) REVERSE BIAS

- p-side connected to -ve terminal and n-side to +ve.
- Potential barrier increases.
- Depletion width increases.
- Majority carriers cannot cross.
- Only small reverse saturation current flows.



3. V-I CHARACTERISTICS OF p-n JUNCTION DIODE



Key Points

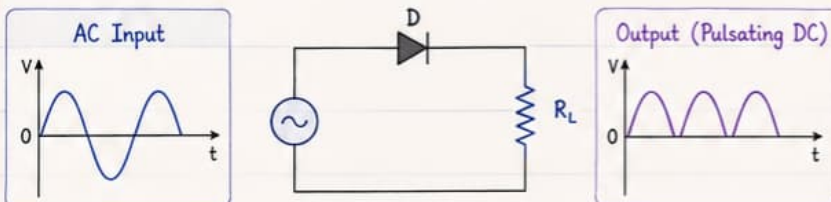
- In forward bias, current increases sharply after knee voltage (V_k).
- In reverse bias, small nearly constant current flows (reverse saturation current).
- At breakdown voltage (V_B), reverse current increases suddenly.

Typical Values

- Knee voltage (V_k) → 0.7 V (Si) ~ 0.3 V (Ge)
- Reverse saturation current (I_s) ~ 10^{-6} A (Si) ~ 10^{-5} A (Ge)

4. DIODE AS A RECTIFIER

- A diode allows current to flow in one direction only. Hence it is used to **convert AC into pulsating DC**.



IDEAL DIODE

- Forward bias → Acts as short circuit ($V_D = 0$, $R_D = 0$)
- Reverse bias → Acts as open circuit ($I_D = 0$, $R_D = \infty$)

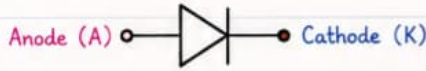
USES OF DIODE

- Rectification
- Signal detection
- Clipping and clamping circuits
- Voltage regulation (Zener diode)
- Electronic switches

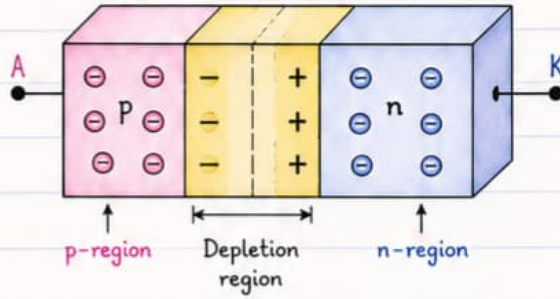
★ SEMICONDUCTOR DEVICES

1. DIODE

- A p-n junction with two terminals is called a **diode**.
- Allows current in **forward bias**, blocks in **reverse bias**.
- Symbol of diode :



Structure of p-n Junction Diode



- ⊖ = Hole
- ⊖ = Electron
- ⊕ = Donor ion
- ⊖ = Acceptor ion

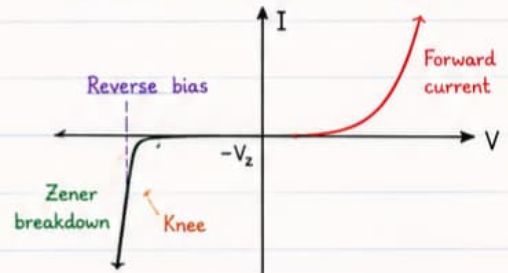
2. ZENER DIODE

- A heavily doped p-n junction diode designed to operate in **reverse bias breakdown region**.
- It maintains a constant voltage (V_Z) across the diode.
- Used for voltage regulation.

Symbol of Zener Diode



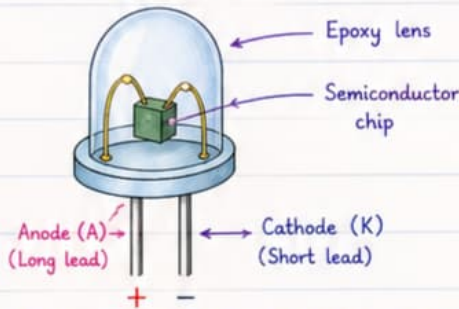
V-I Characteristics (Zener Diode)



3. LED (LIGHT EMITTING DIODE)

- A p-n junction diode made of **direct band gap semiconductor**.
- When forward biased, electrons and holes recombine and **emit light**.
- Emits light of specific colour depending on the material used.

Structure of LED



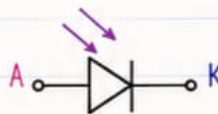
Key Points

- LED works only in forward bias.
- No current in reverse bias (ideally).
- Used in indicators, displays, lighting, optical communication etc.

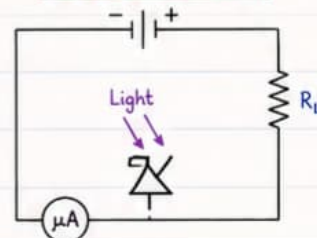
4. PHOTODIODE

- A p-n junction diode designed to work in **reverse bias**.
- When light falls on the junction, **electron-hole pairs** are generated.
- This **increases** the **reverse current**, which is proportional to **intensity of light**.
- Used in light detectors, cameras, optical communication.

Photodiode Symbol



Reverse Bias Connection



Note

More light
→ more reverse current.
Hence, current is a measure of light intensity.

IMPORTANT COMPARISON

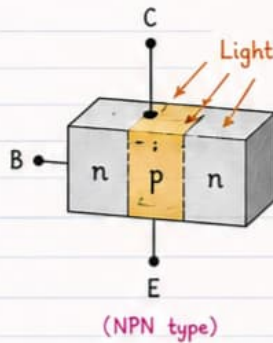
Device	Bias	Main Function	Key Feature	Applications
Diode	Forward bias (conducts)	Allows current in one direction	Rectification	Power supplies, Rectifiers
Zener Diode	Reverse bias (breakdown)	Maintains constant voltage (V_Z)	Voltage regulation	Voltage regulators, Reference voltage
LED	Forward bias	Emits light	Light emission	Indicators, Displays, Lighting
Photodiode	Reverse bias	Detects light	Light → current conversion	Light sensors, Cameras, Optical communication

★ Remember : Understanding diodes and their applications is the key to modern electronics!

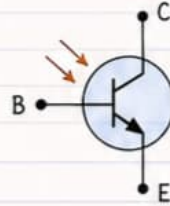
SEMICONDUCTOR DEVICES (Contd.)

5. PHOTOTRANSISTOR

- A transistor whose base region is kept **light sensitive**.
- Light falling on base reduces its resistance and increases **collector current**.
- Used in light sensing circuits.



Phototransistor Symbol

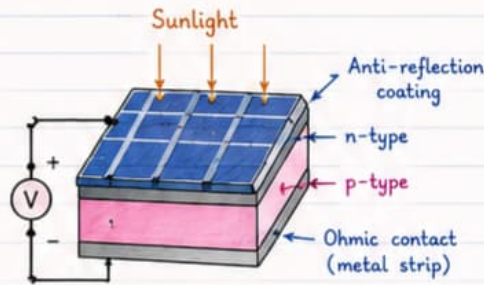


Key Points

- Works in reverse bias.
- Incident light increases collector current.
- Used in automatic street lights, camera exposure meters, alarms.

6. SOLAR CELL (PHOTOVOLTAIC CELL)

- A p-n junction diode with large area which converts **light energy** into **electrical energy**.
- When light falls on the junction, emf is developed across it (**photovoltaic effect**).
- Used to generate DC voltage.
- Pollution free and renewable source of energy.



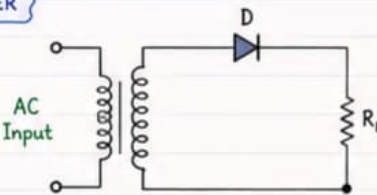
Features

- No moving parts
- Low maintenance
- Long life
- Used in satellites, calculators, solar chargers, street lights

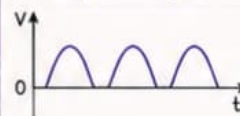
★ SIMPLE ELECTRONIC CIRCUITS (Using Semiconductor Devices)

7. HALF WAVE RECTIFIER

- Uses one diode.
- Allows current in one half cycle only.
- Output is pulsating DC.



Output Waveform



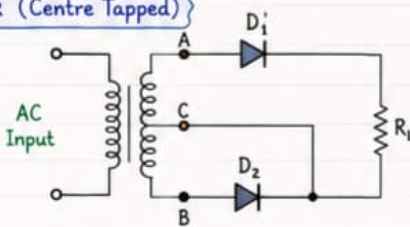
Applications

- Battery charging
- Power supplies
- Signal detection

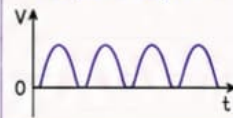


8. FULL WAVE RECTIFIER (Centre Tapped)

- Uses two diodes.
- Allows current in both half cycles.
- Output is pulsating DC with **higher efficiency**.



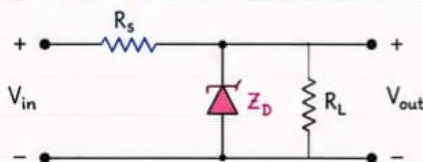
Output Waveform



Advantages over Half Wave

- ✓ Uses full cycle of input.
- ✓ Higher output voltage.
- ✓ More efficient.
- ✓ Less ripple.

9. ZENER DIODE AS VOLTAGE REGULATOR



- Zener diode is reverse biased in breakdown region.
- Voltage across it remains constant ($= V_Z$) even if input voltage or load resistance changes.
- Used in **voltage regulators**.

Condition for Regulation

$$I_{Z(\min)} \leq I_Z \leq I_{Z(\max)}$$

Where,

$I_{Z(\min)}$ = Minimum Zener current

$I_{Z(\max)}$ = Maximum Zener current

★ QUICK SUMMARY

- Intrinsic semiconductor → Pure Si/Ge, both electrons & holes.
- Extrinsic semiconductor → Doping creates majority carriers.
- Diode → Allows current in one direction.
- Zener diode → Reverse bias breakdown for voltage regulation.
- LED → Emits light in forward bias.
- Photodiode / Phototransistor → Detect light.
- Solar cell → Converts light energy into electrical energy.
- Rectifiers → Convert AC into DC.

Semiconductor devices are the building blocks of modern electronics. They make possible devices like computers, phones, LEDs, solar panels and more! ❤️

NEET TIP

- ★ Understand working principle with energy band diagram.
- ★ Learn symbols and V-I characteristics well.
- ★ Practice numericals on circuits and diode applications.